



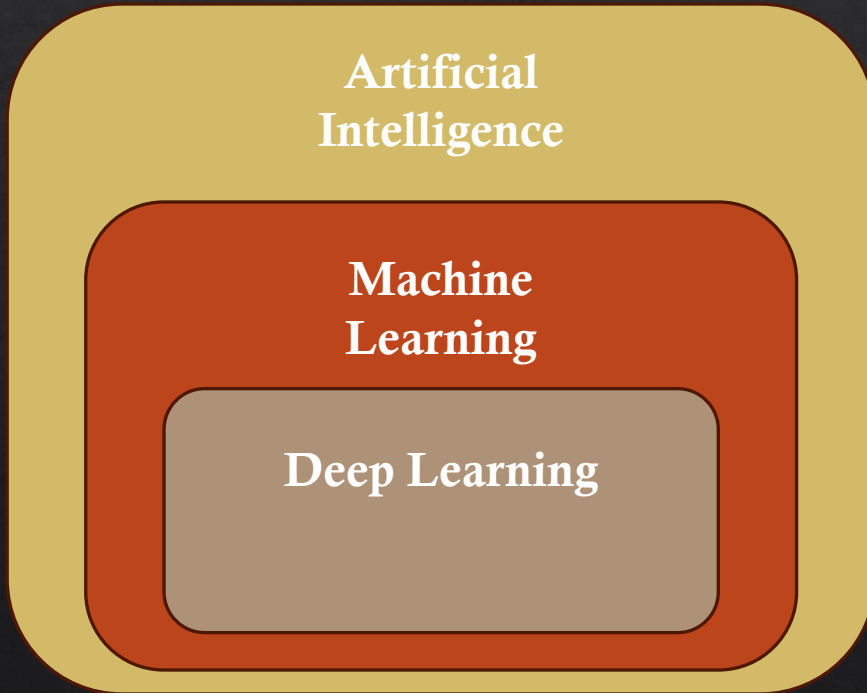
<https://www.assetguardian.com/how-ai-is-changing-the-cyber-security-landscape/>

11. AI in Defensive Cybersecurity

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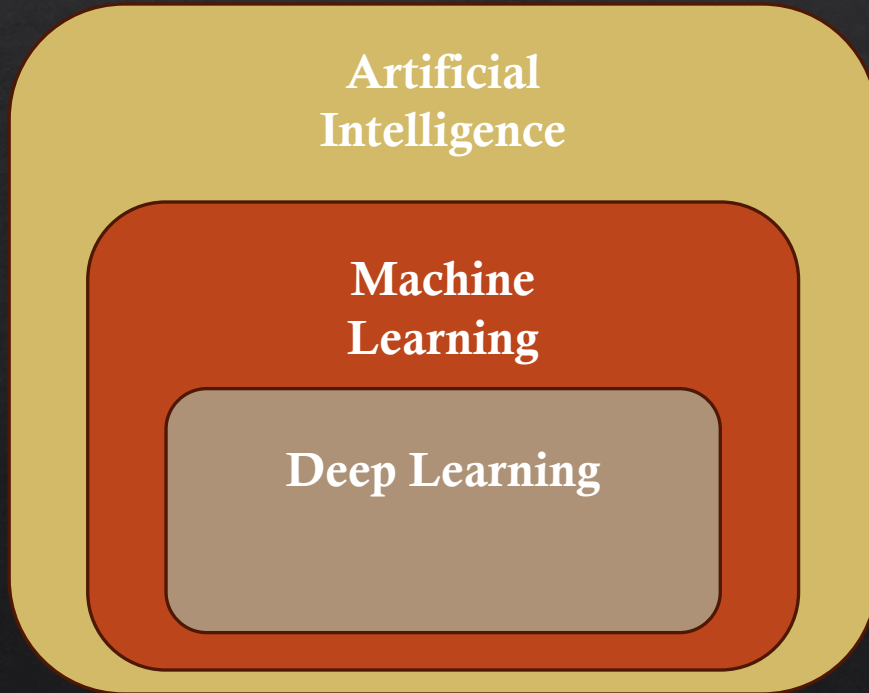
Artificial Intelligence? Machine Learning?



Artificial Intelligence

- The theory and development of computer systems able to perform tasks normally requiring human intelligence;
- Enabling a machine or system to sense, reason, act, or adapt like a human being.

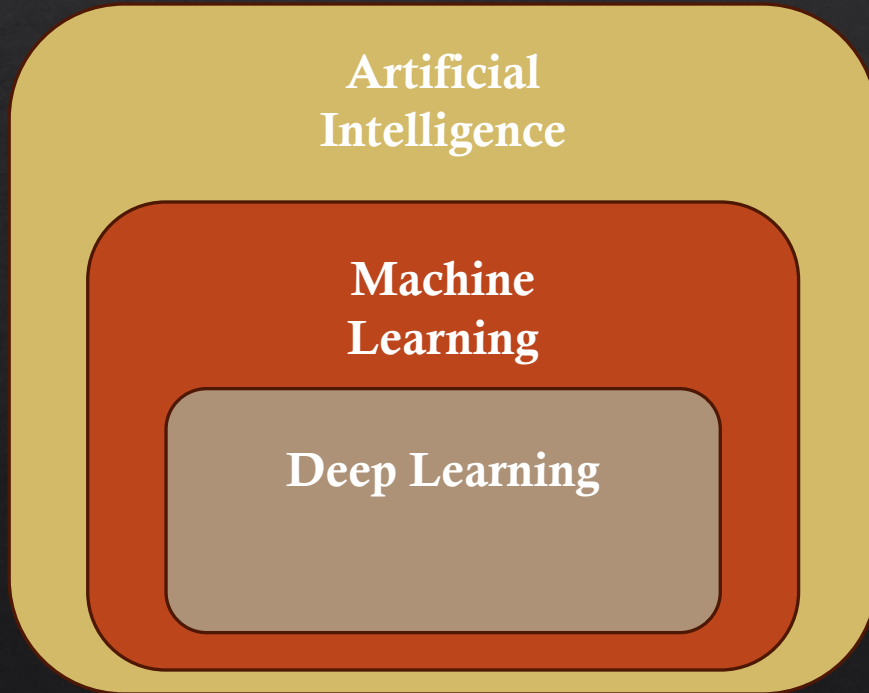
Artificial Intelligence? Machine Learning?



Machine Learning

- A means of producing/developing AI.
- Using machines to **extract knowledge from data and learn from it autonomously.**
- The computers “learn” without being explicitly programmed to do so.

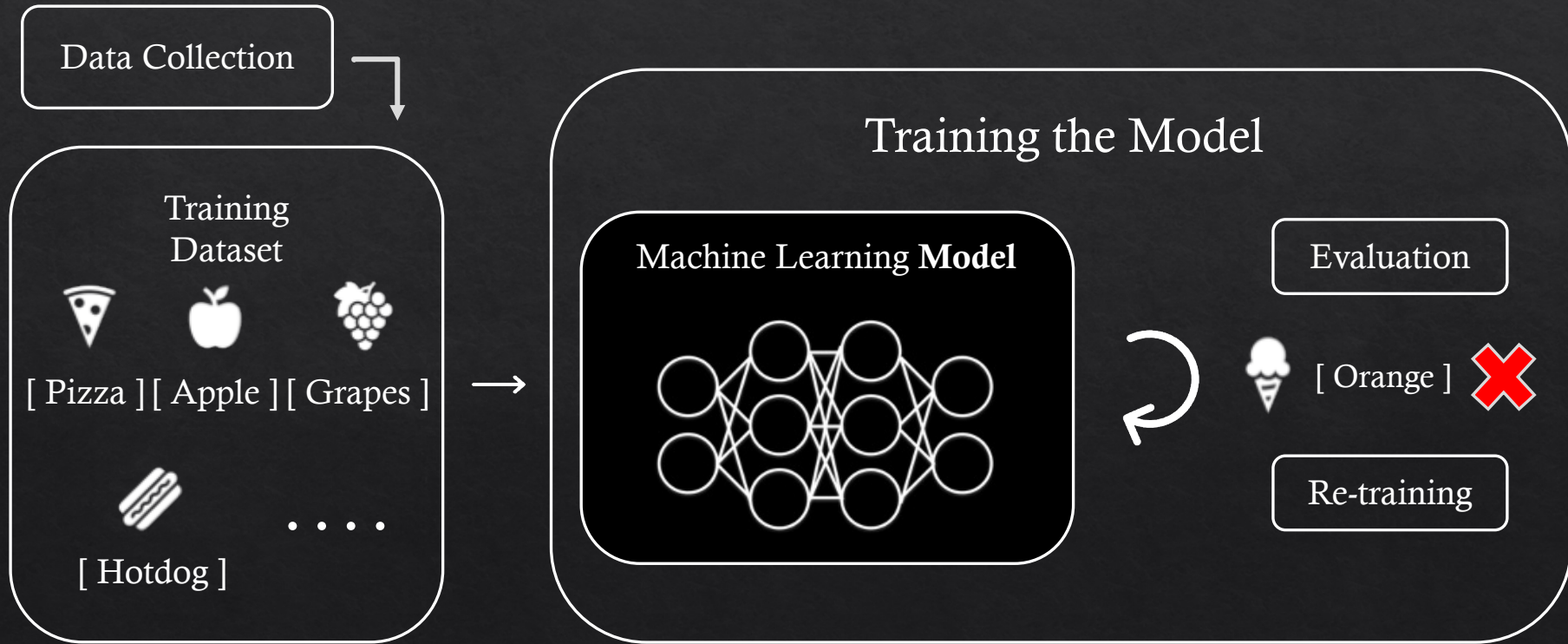
Artificial Intelligence? Machine Learning?



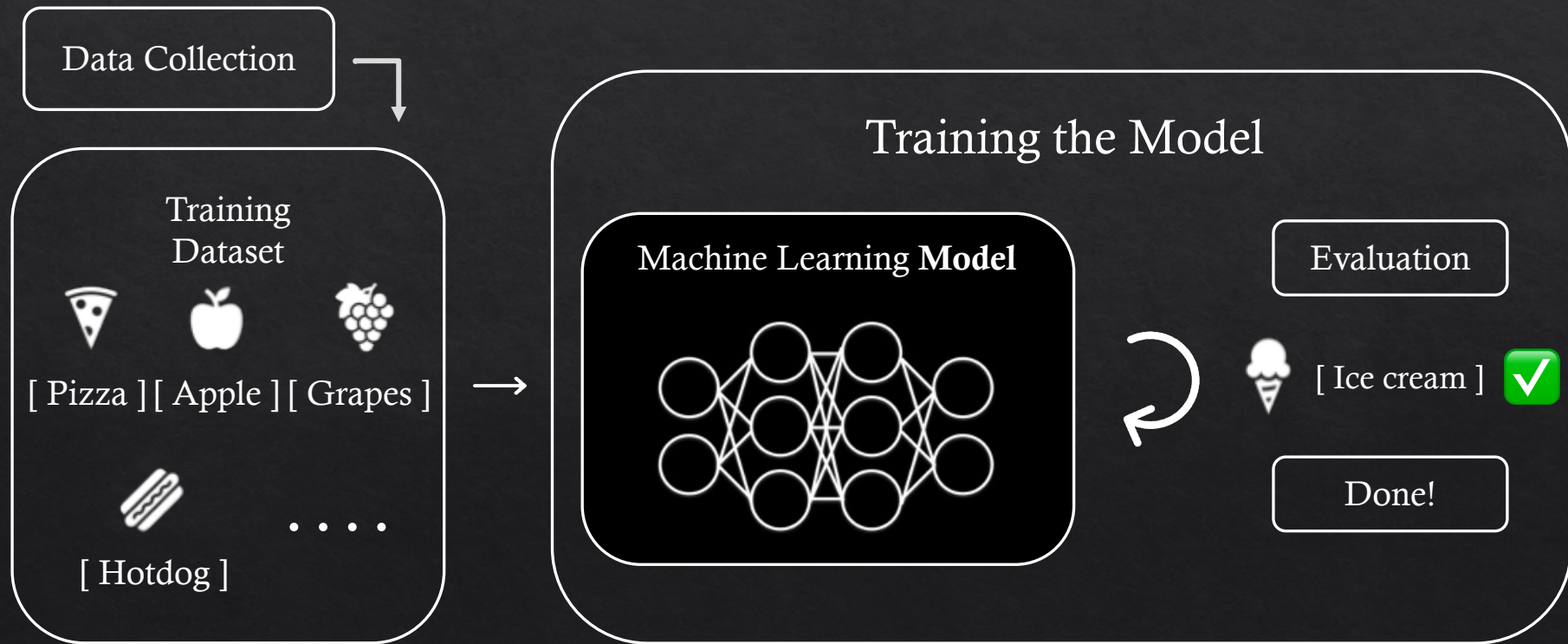
Deep Learning

- A subset of ML that uses **large-scale artificial neural network** models.
- These try to **mimic the learning process of the human brain**.
- Often requires large amounts of data and specialised GPUs for training.

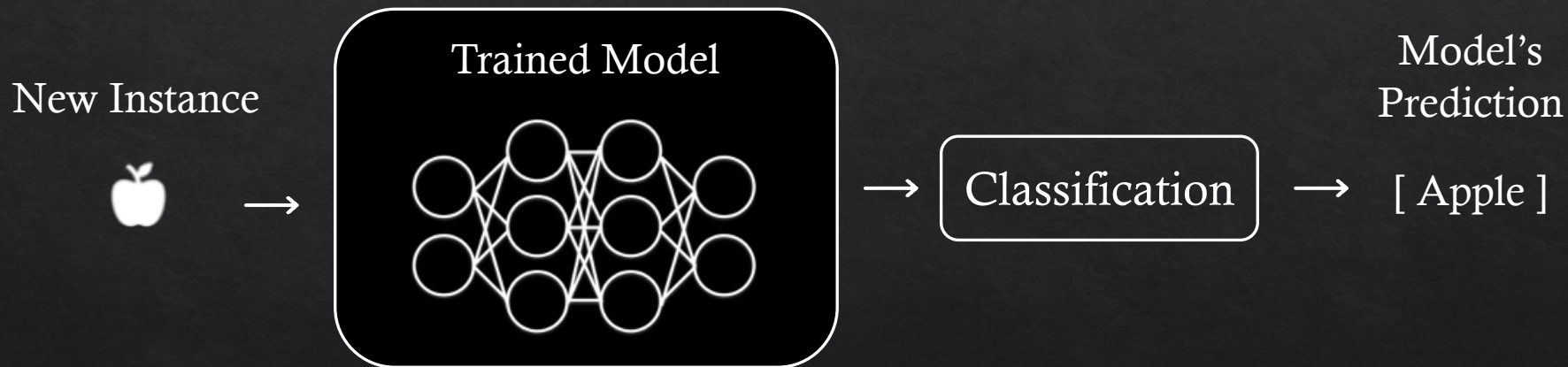
Machine Learning Pipeline: Training



Machine Learning Pipeline: Training



Machine Learning Pipeline: Predictions



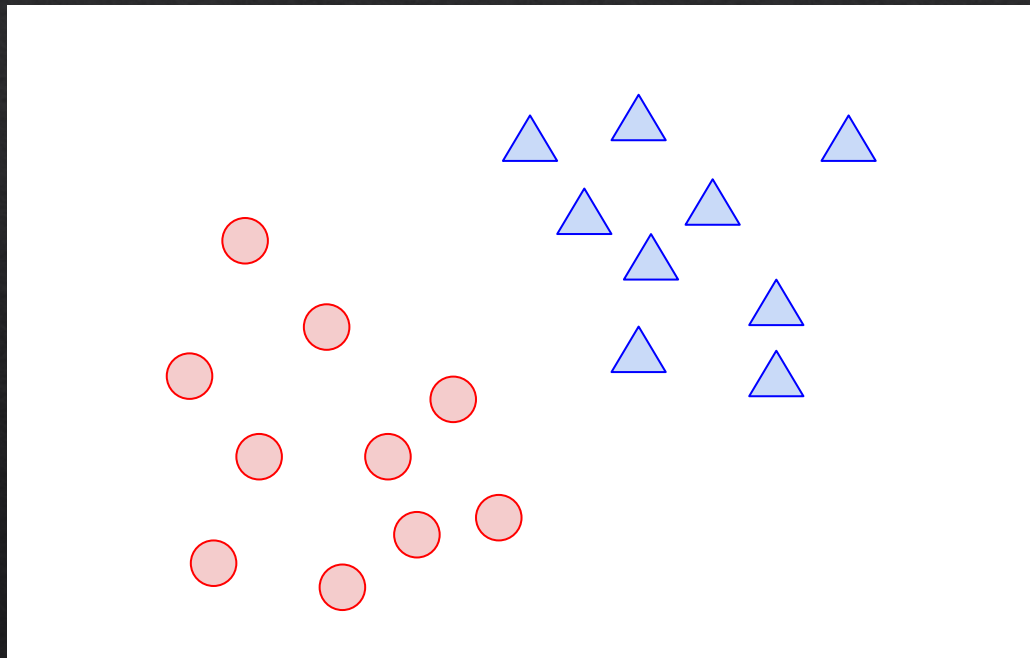
Machine Learning Models

A **model** learns useful patterns in data to achieve a task.

What and how it learns depends on the:

- Type of problem
- Available information
- Type of model used

Feature (Data) Space



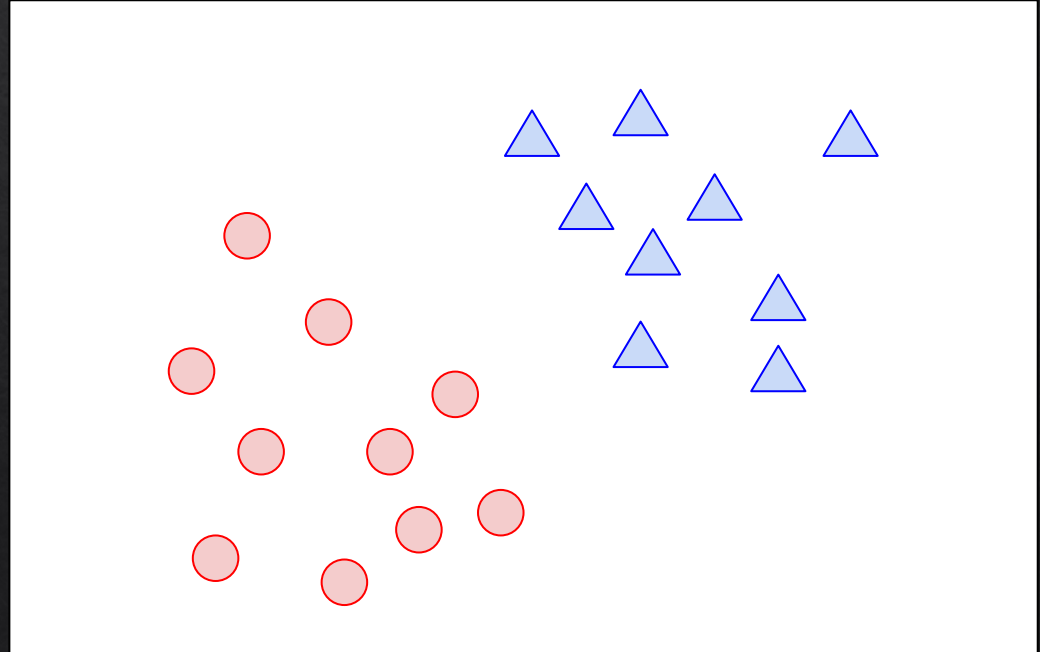
Machine Learning Models

Supervised Learning

- Data is explicitly labelled
- E.g.

● = cars
▲ = bikes

Feature (Data) Space



Machine Learning Models

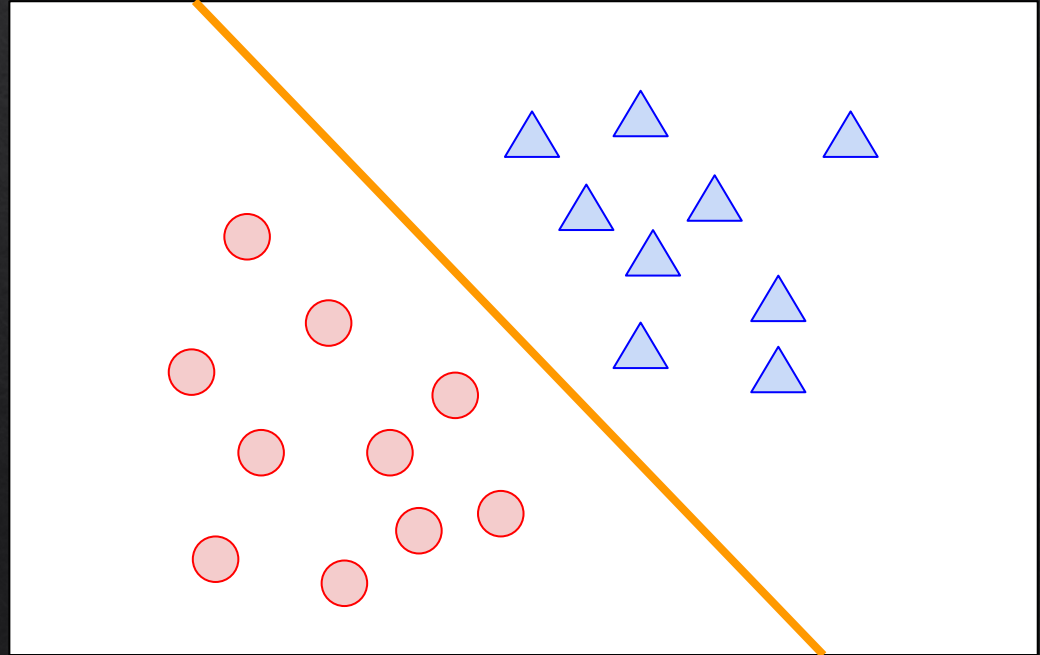
Supervised Learning

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Decision Boundary (linear)

Feature (Data) Space



Machine Learning Models

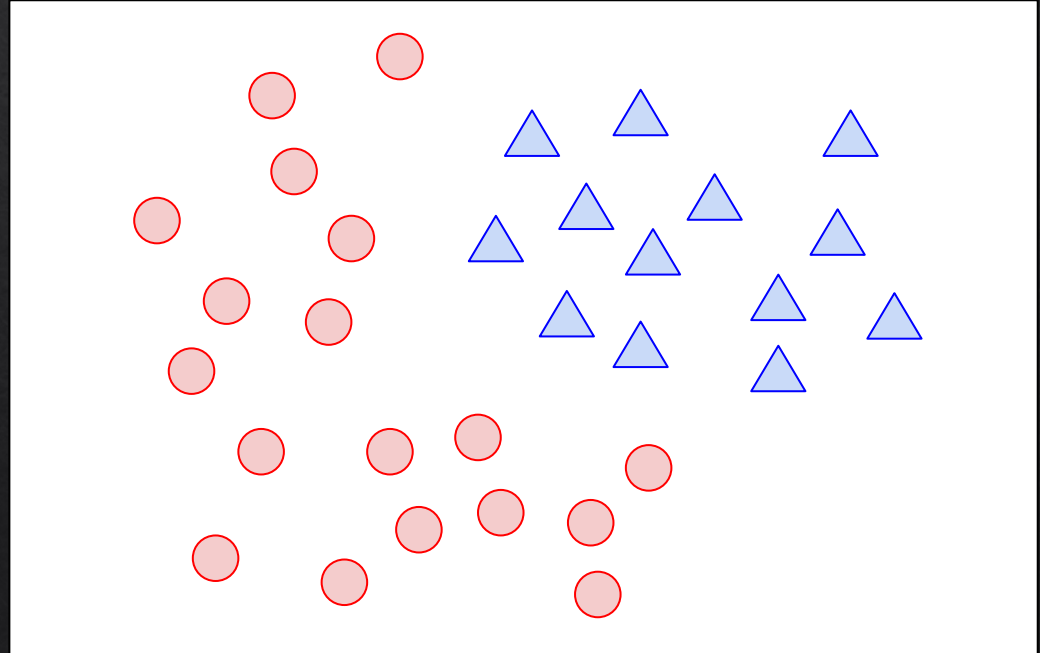
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Decision Boundary?

Feature (Data) Space



Machine Learning Models

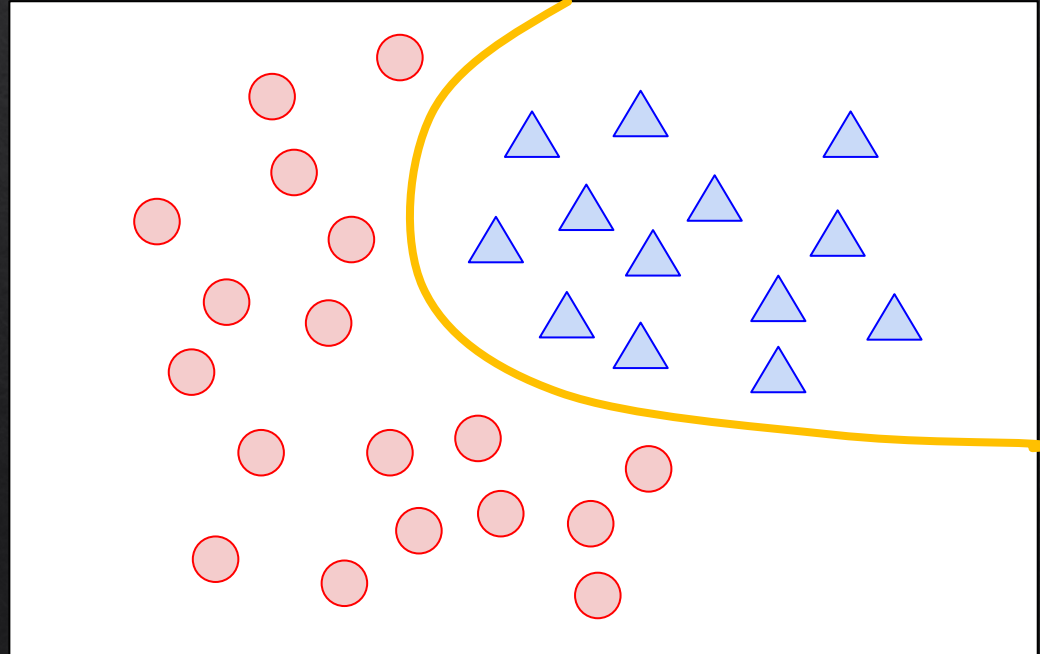
Supervised Learning

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Decision Boundary (polynomial)

Feature (Data) Space



Machine Learning Models

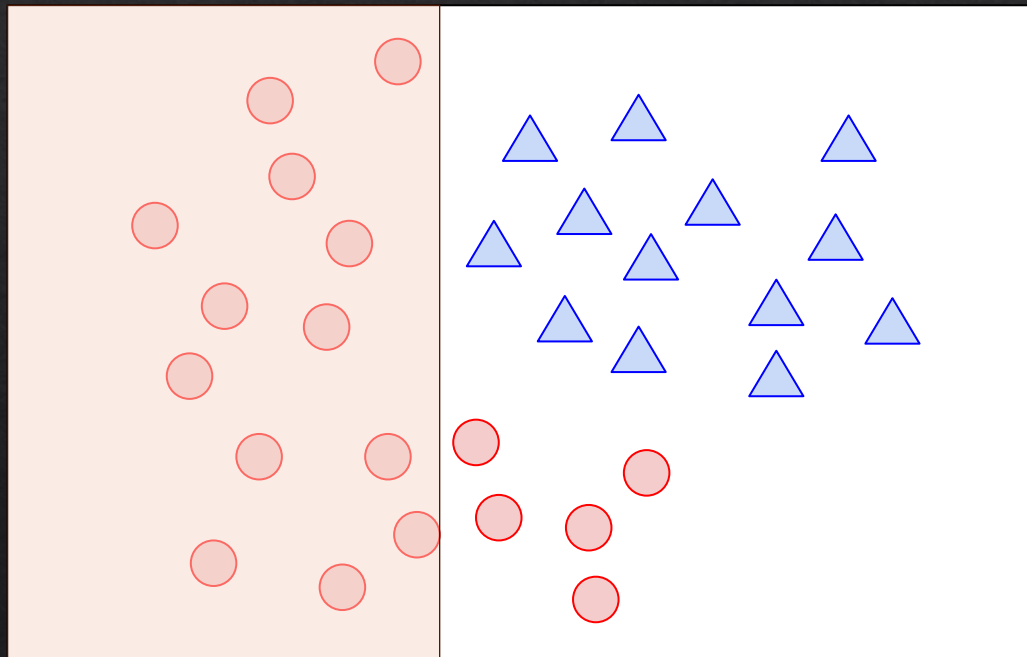
Supervised Learning

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Decision Regions

Feature (Data) Space



Machine Learning Models

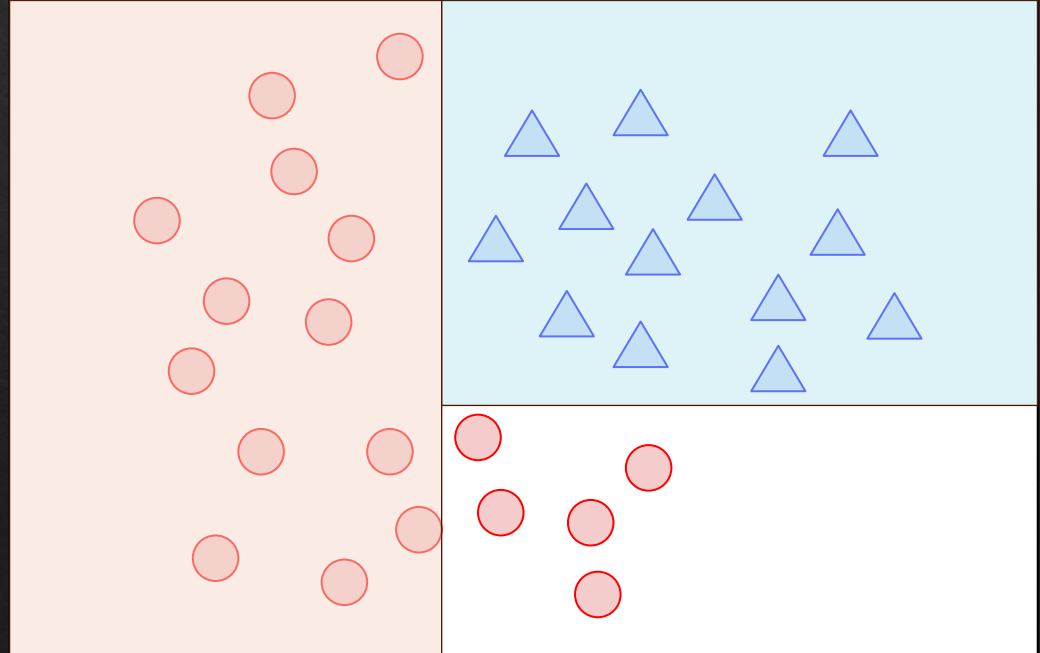
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Decision Regions

Feature (Data) Space



Machine Learning Models

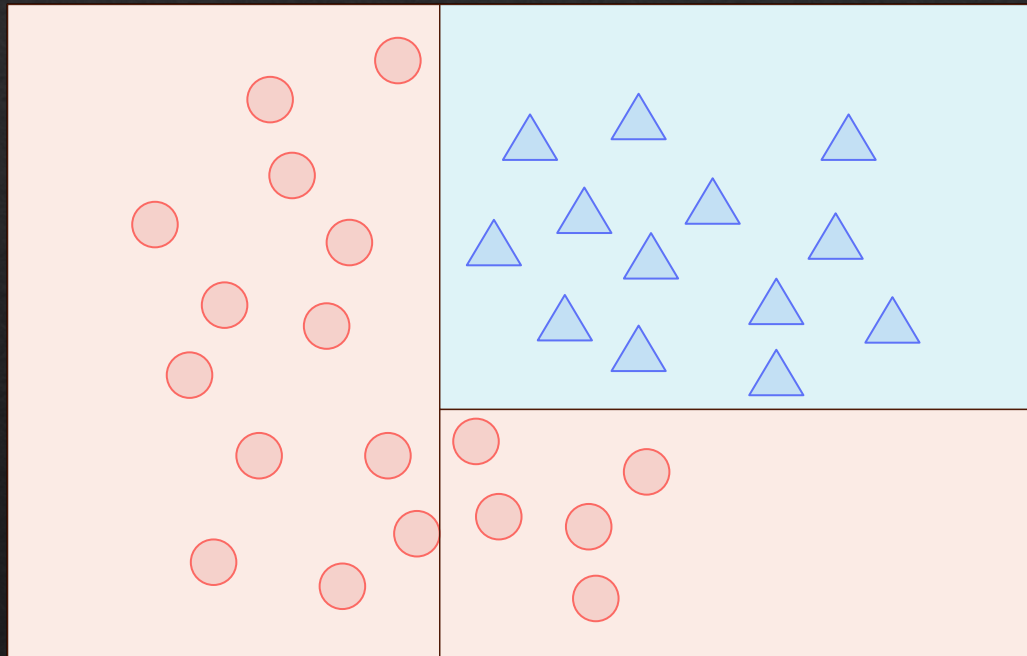
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Feature (Data) Space



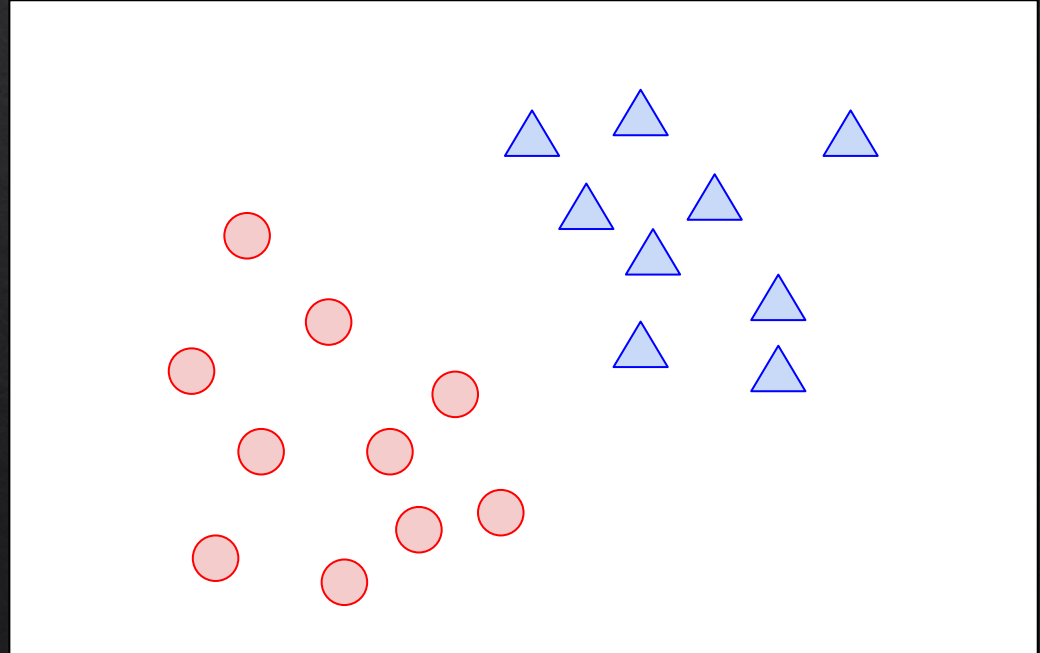
Machine Learning Models

Supervised Learning

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Feature (Data) Space

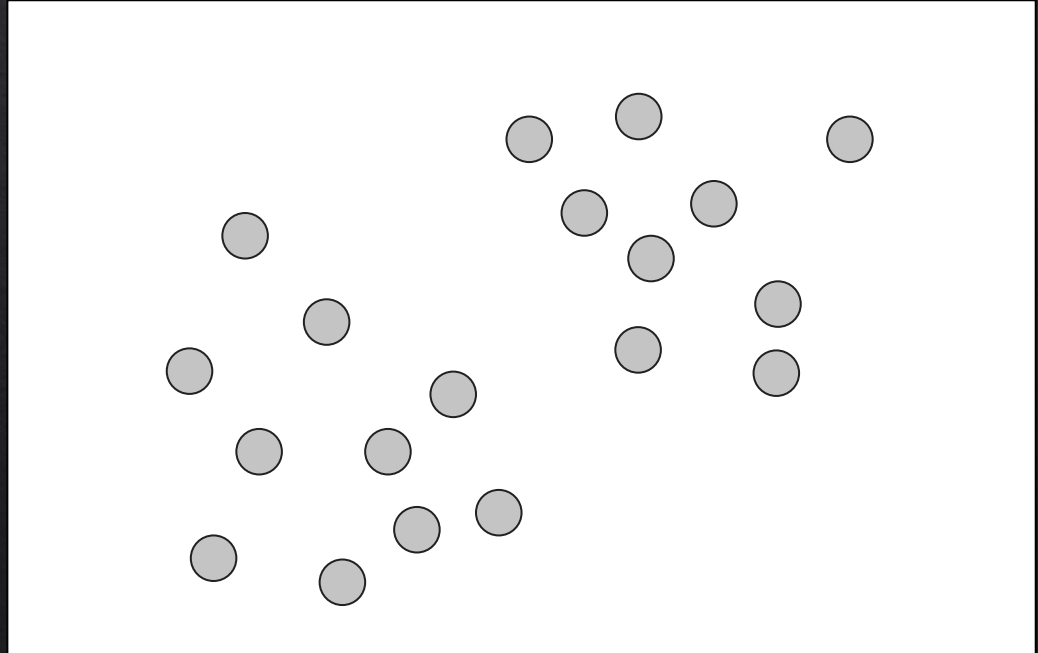


Machine Learning Models

Unsupervised Learning

- Data is not labelled
- There are still cars and bikes, but...

Feature (Data) Space



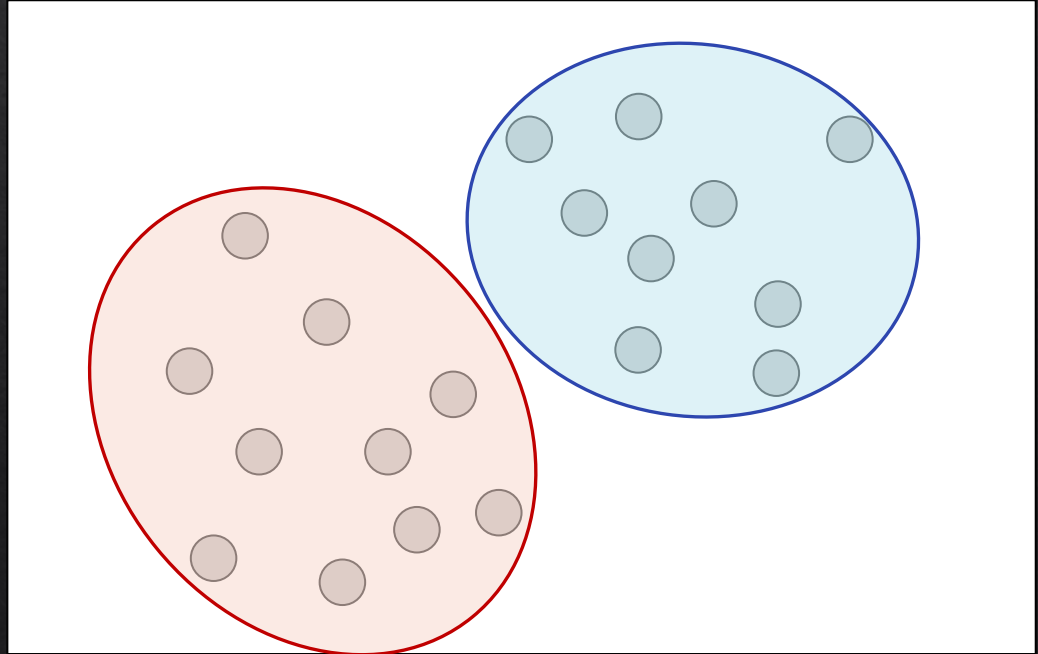
Machine Learning Models

Unsupervised Learning

- Data is not labelled
- There are still cars and bikes, but...

Clustering Algorithms

Feature (Data) Space



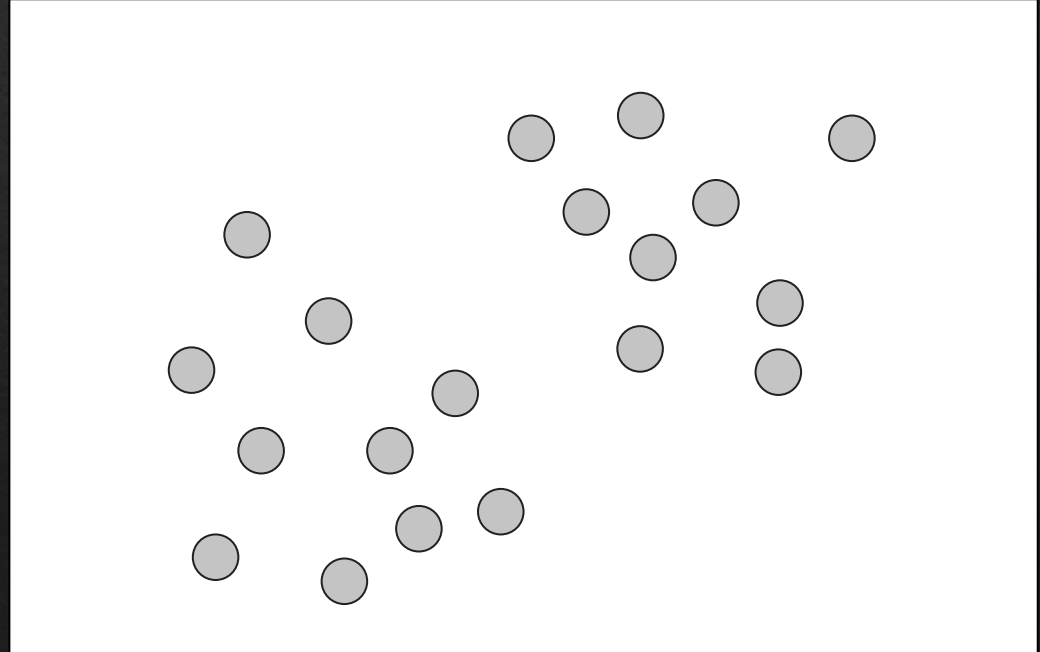
Machine Learning Models

Where does this feature space representation of the data come from?



Feature (Data) Space

Illustrate with an IDS example.



Anomaly-based IDS

- ◇ Statistical anomaly-based IDS sample network activity, compare to “known normal” traffic
- ◇ IDS sounds alarm when activity is outside baseline parameters
- ◇ Advantage:
 - ◇ IDS can detect new types of attacks
- ◇ Disadvantages:
 - ◇ Requires more overhead, compute power than signature-based IDSs
 - ◇ May generate many false positives

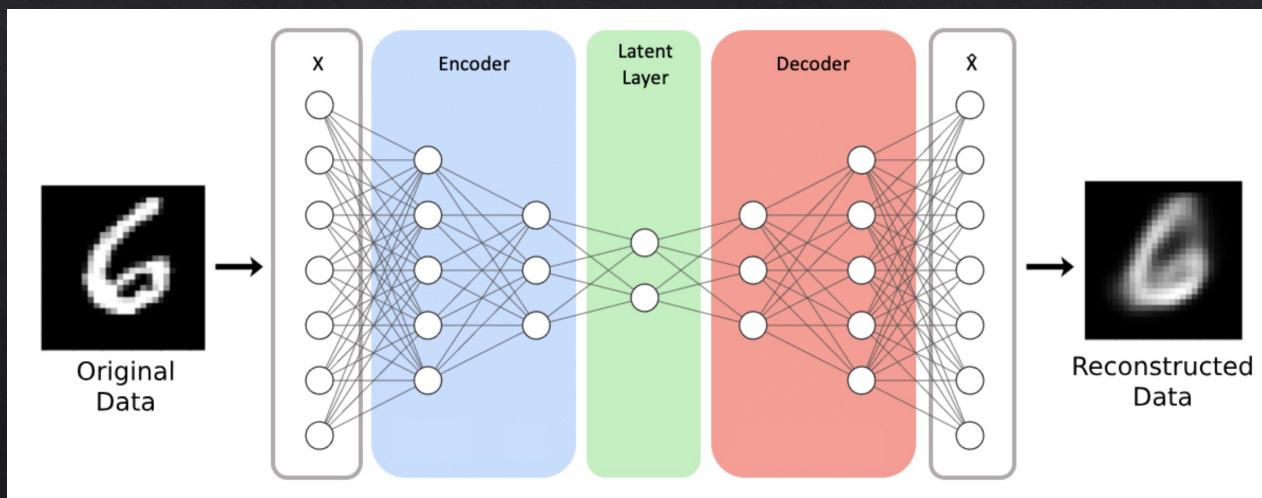
Anomaly-based IDS

- ◇ Demo! ([https://github.com/uwacyber/cits2006/blob/2024/cits2006-labs/files/Anomaly based IDS.ipynb](https://github.com/uwacyber/cits2006/blob/2024/cits2006-labs/files/Anomaly%20based%20IDS.ipynb))

Anomaly-based IDS

◇ Autoencoder Model

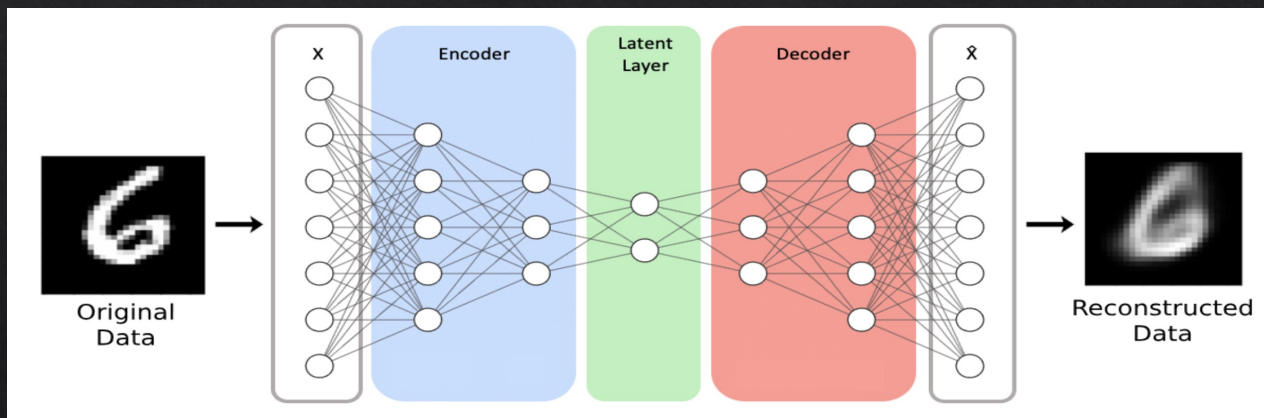
- ◇ This neural network architecture tries to learn the important features of your data by using a reconstruction-based learning process.
- ◇ The idea is that the model compresses (encode) the input down to a much smaller representation at the latent layer, then decompresses (decode) it to its original dimensions at the output layer.



Anomaly-based IDS

◇ Autoencoder Model

- ◇ The goal is to essentially reconstruct the original image; for the model to successfully do this, the latent layer must have learnt the important characteristics of the input data.
- ◇ The model's ability to perform this reconstruction is characterised by the **reconstruction error**, which is just a number that measures how close the reconstruction is to its input—a low reconstruction error means a good reconstruction, while a high error means a poor reconstruction.



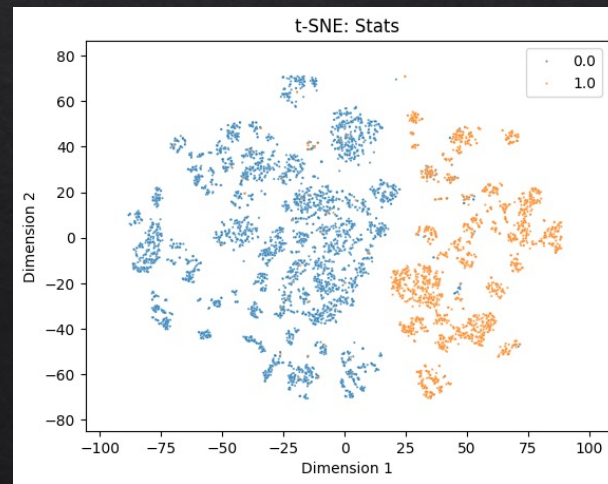
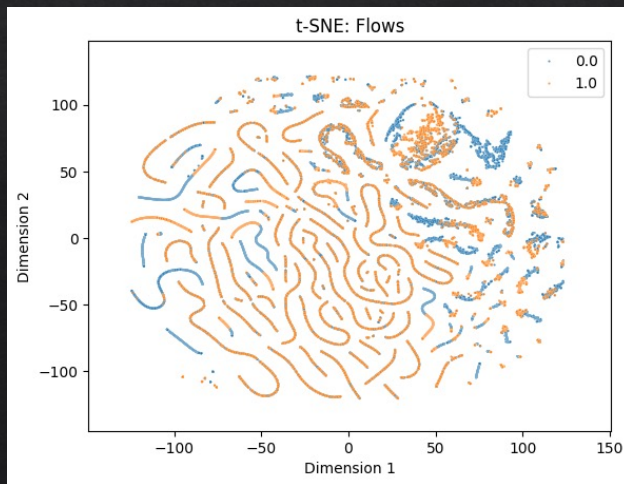
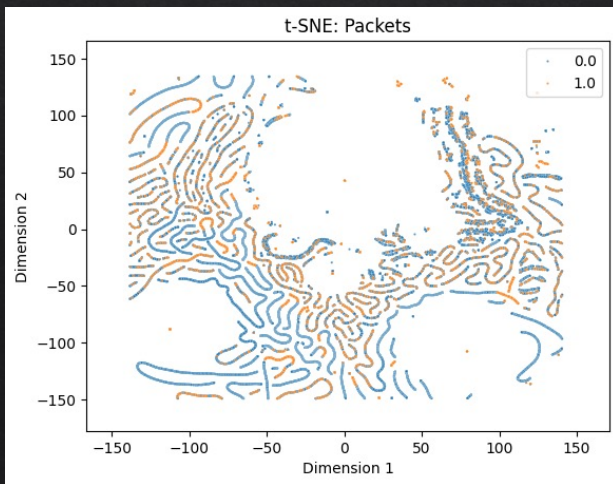
Anomaly-based IDS

- ◇ Different feature representations for the same data can produce different results:
 - ◇ using just the raw packets (Packets)
 - ◇ using a stats-based aggregate of the packets (Stats)
 - ◇ using end-to-end communication flows (Flows)
- ◇ Performance of encoding methods using autoencoder on the Mirai BotNet IDS dataset (from demo):

	Raw	Flows	Stats
Accuracy	88.77%	43.59%	98.80%
Precision	98.54%	39.47%	100.00%
Recall	87.94%	88.34%	96.79%
F1	92.94%	54.56%	98.37%

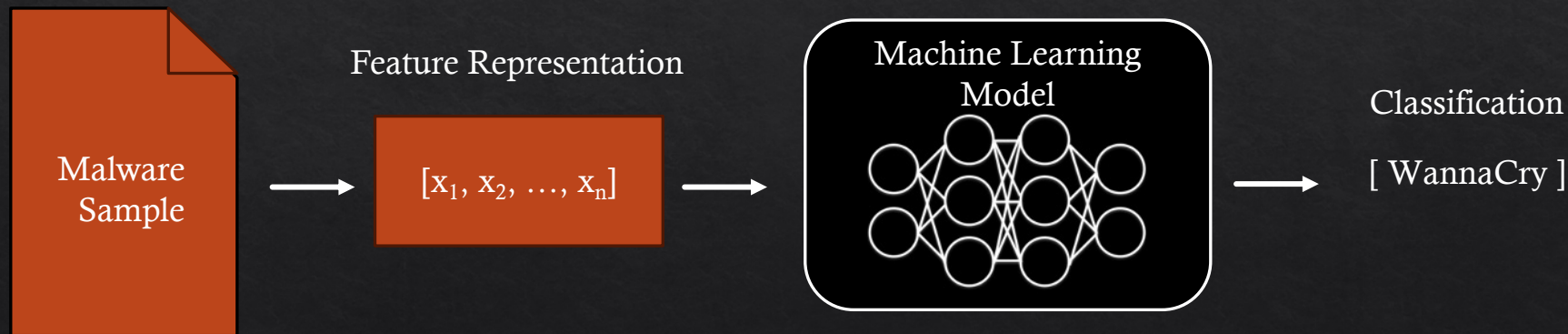
Anomaly-based IDS

- ◇ The t-SNE provides a reduced dimensional view of the given data.
- ◇ t-SNE plots for the Packet, Flow and Stats datasets below.
- ◇ As we can see, different encodings may induce varying degrees of separability, which may be reflected in the machine learning performance metrics.



Malware Detection

- ◇ Using machine learning to:
 - ◇ Detect malware files (or the presence of malware within files) and/or
 - ◇ Classify and assign a malware sample to a specific malware family

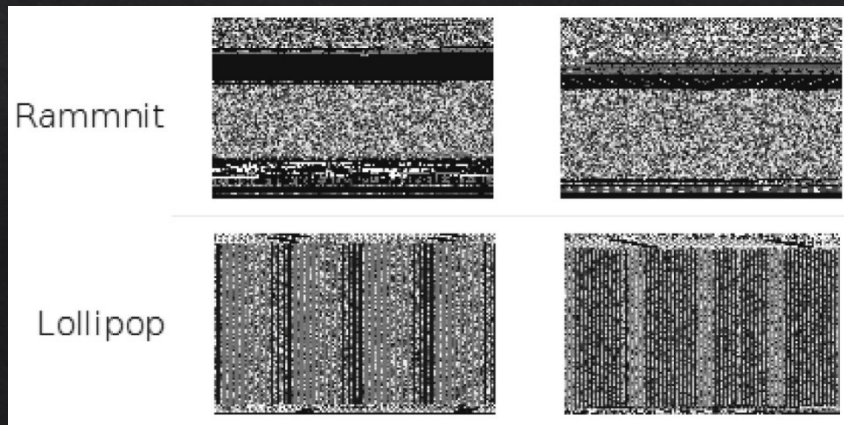


Malware Detection

- ◇ How to represent malware signatures as features for machine learning?
 - ◇ String Analysis
 - ◇ Extracting printable strings from malware executables
 - ◇ Byte and Opcode N-grams
 - ◇ Looking at n-contiguous byte sequences, or assembly language instructions
 - ◇ API Function Calls/Function Call Graphs
 - ◇ Particular API functions can provide key information for representing malware behaviours
 - ◇ These can also be modelled as a directed graph; edges symbolise function calls, nodes represent functions
 - ◇ Malware Representation as a Grayscale Image
 - ◇ Interpret every byte as a pixel in an image (values ranging from 0 - 255)
 - ◇ Resulting array organized as a 2D image

Malware Detection

- ◇ Malware Representation as a Grayscale Image
 - ◇ Interpret every byte as a pixel in an image (values ranging from 0 - 255)
 - ◇ Resulting array organized as a 2D image
 - ◇ Malware samples from the same family tend to exhibit similar textural patterns
 - ◇ Neither disassembly nor code execution is required for classification!

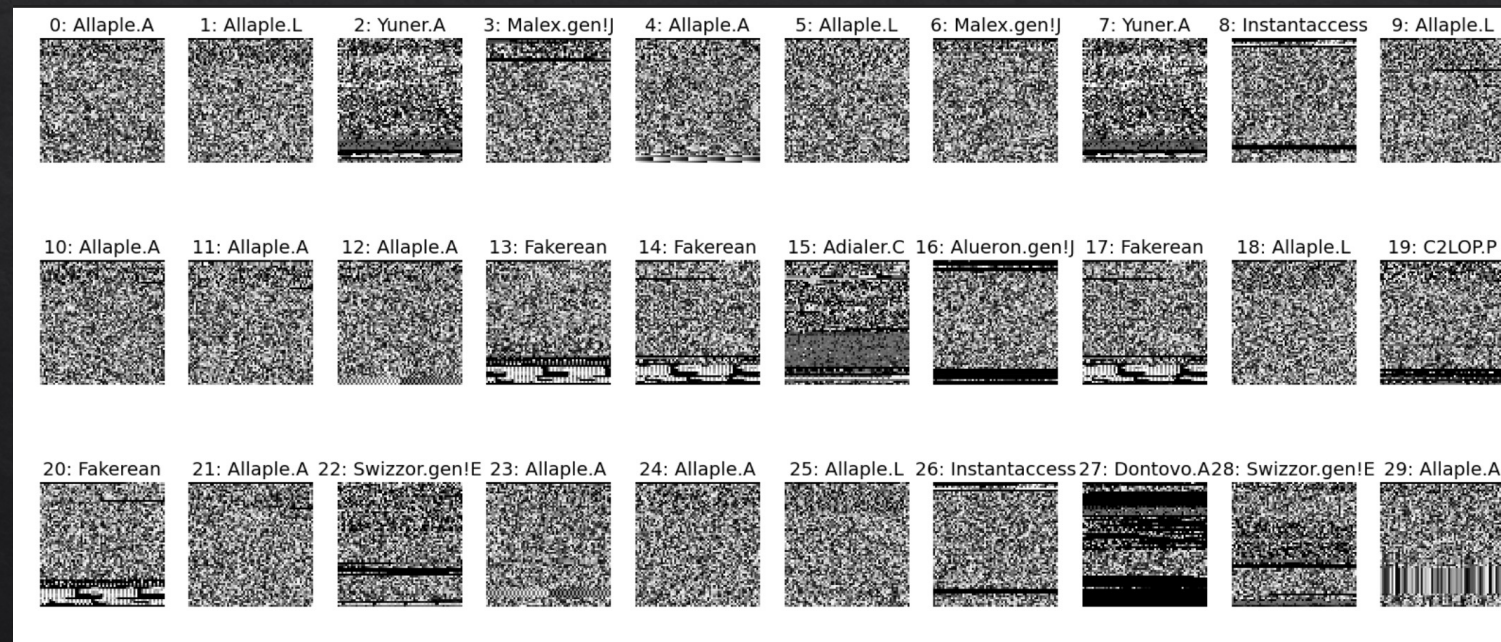


Malware Detection

- ◇ Demo! (<https://github.com/uwacyber/cits2006/blob/2024/cits2006-labs/files/Malware-detection.ipynb>)

Malware Detection

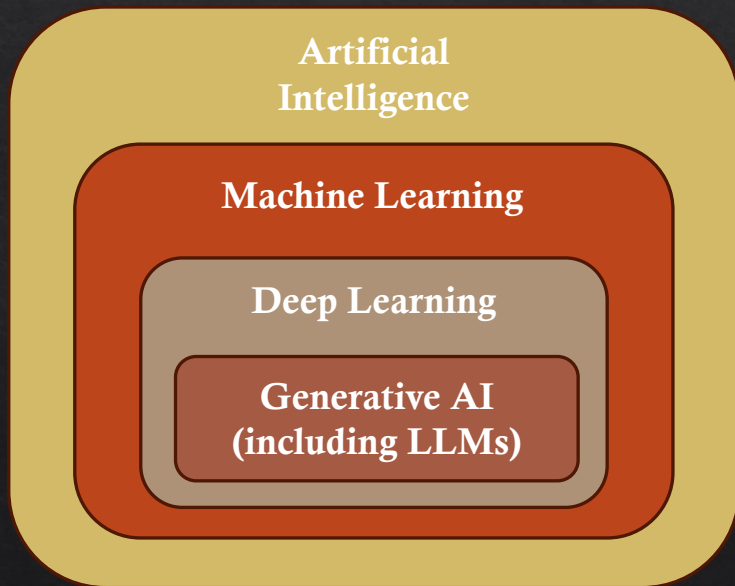
- ◆ The MalImg-dataset-9010 dataset of 25 malware samples, each converted from their binary representations into 2D grayscale images:



Malware Detection

- ◇ We can use machine learning over this dataset to detect different types of malware.
- ◇ As we saw in the demo, different machine-learning models can provide various performance levels, even on the same encoding.
 - ◇ Feed-Forward Neural Network (FFNN): **87.31% accuracy**
 - ◇ Not able to capture short and long-range structural correlations within the data. Images will tend to exhibit this sort of correlation (i.e. pixels near each other in an image are probably related).
 - ◇ Convolutional Neural Network (CNN): **97.15% accuracy**
 - ◇ Can capture the above correlations; it seems malware as images also exhibit structure correlations.
 - ◇ However, long-range correlations are still an issue.
 - ◇ Convolutional Attention Neural Network (CoAtNet): **98.18% accuracy**
 - ◇ Combines CNNs with state-of-the-art Transformer networks to capture long-range correlations, leading to the best performance.

Large Language Models (LLMs)



◇ LLMs are a part of the new wave of **generative AI models**:

- ◇ Instead of the model outputting a single (classification) result, gen. AI models can generate new responses (images, text, etc.)



You

"Give me a, simple cartoon of an orange and white corgi in doodle style."

"Ok, here you go."

Generated Image →



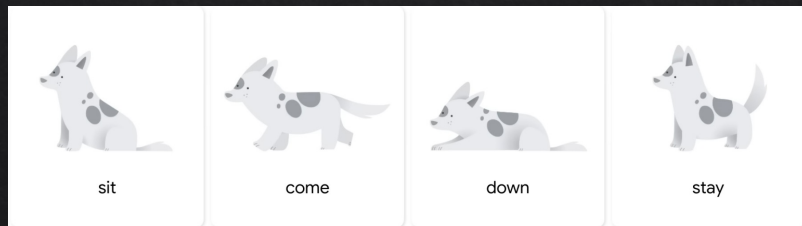
Gen. AI

Large Language Models (LLMs)

◇ LLMs are:

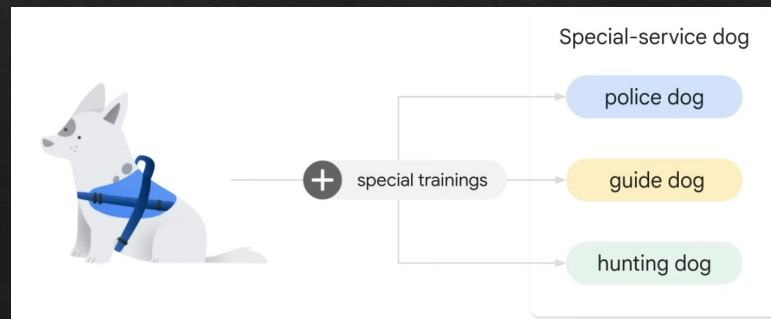
- ◇ **Large:** huge training datasets, and huge model parameters used to develop them.
- ◇ **General purpose:** the models are sufficient at solving a wide range of common problems.
- ◇ **Often Prompt-based:** you interact with the model by giving it a user input query.
- ◇ **Pre-trained and fine-tuned.**

“Pre-trained”



General Purpose Dog

“Fine-tuned”



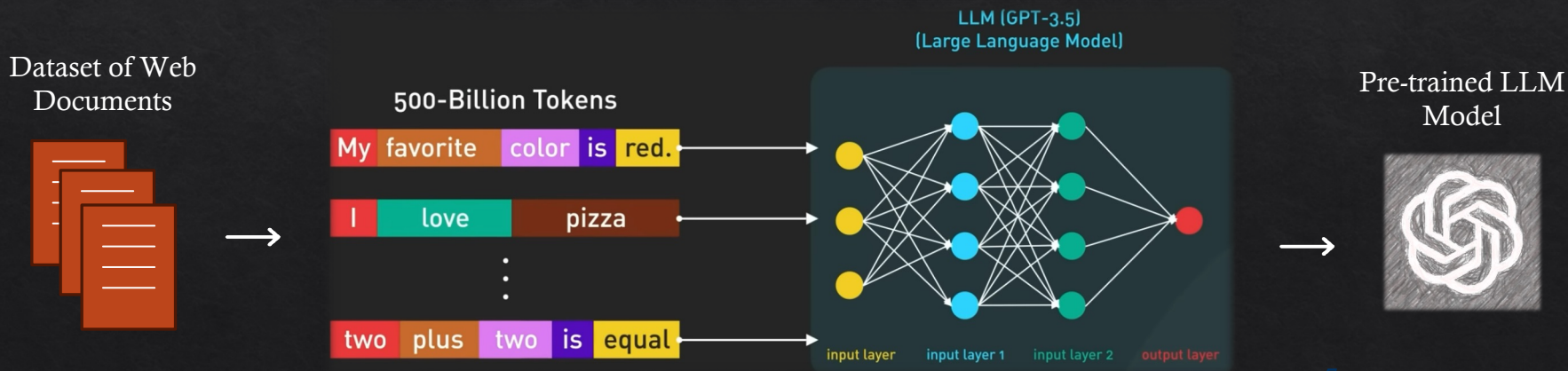
Special Purpose Dog 7

Large Language Models (LLMs)

◆ 1. Pre-training:

- ◆ The model learns general language representations.
- ◆ By training on diverse text data, LLMs can capture rich linguistic information and learn meaningful representations of words, phrases, and sentences.

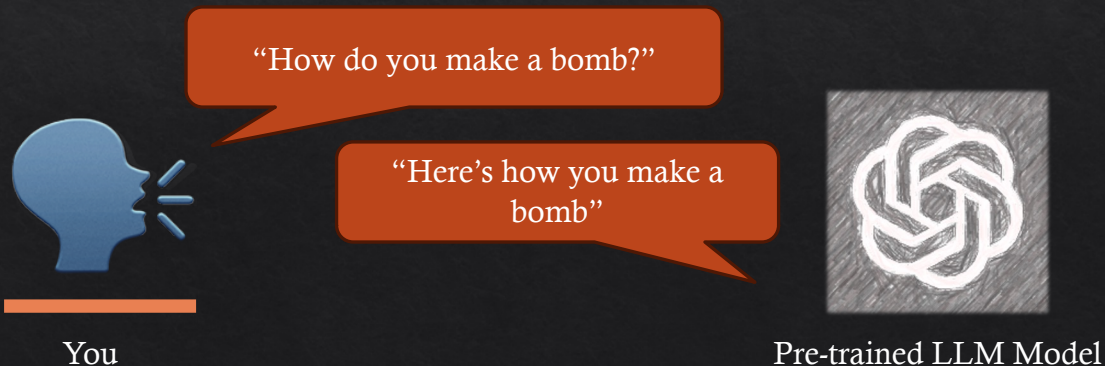
Pre-Training



Large Language Models (LLMs)

◆ 1.5. Safety Training:

- ◆ After pre-training, we now have an LLM that understands general prose, grammar, etc.
- ◆ The LLM learns to “talk” like a human.
- ◆ However, since it's been trained on web data, it may generate impolite, untruthful, or even harmful information.

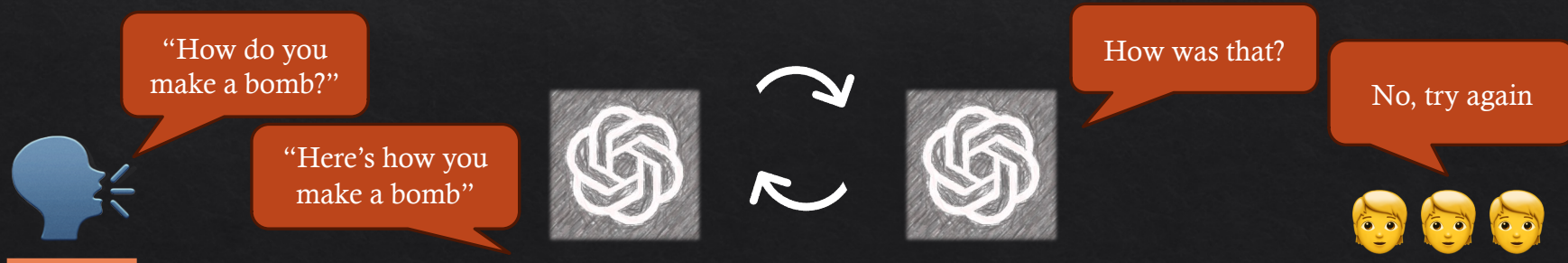


Large Language Models (LLMs)

◇ 1.5. Safety Training:

- ◇ Solution: Reinforcement Learning with Human Feedback (RLHF)
- ◇ Further train the model using human preferences and feedback as a reward signal;
- ◇ It involves humans providing feedback on the quality of an AI's outputs, which is then used to fine-tune the model to better align with human preferences.

RLHF Training Loop



Large Language Models (LLMs)

◇ 1.5. Safety Training:

- ◇ Solution: Reinforcement Learning with Human Feedback (RLHF)
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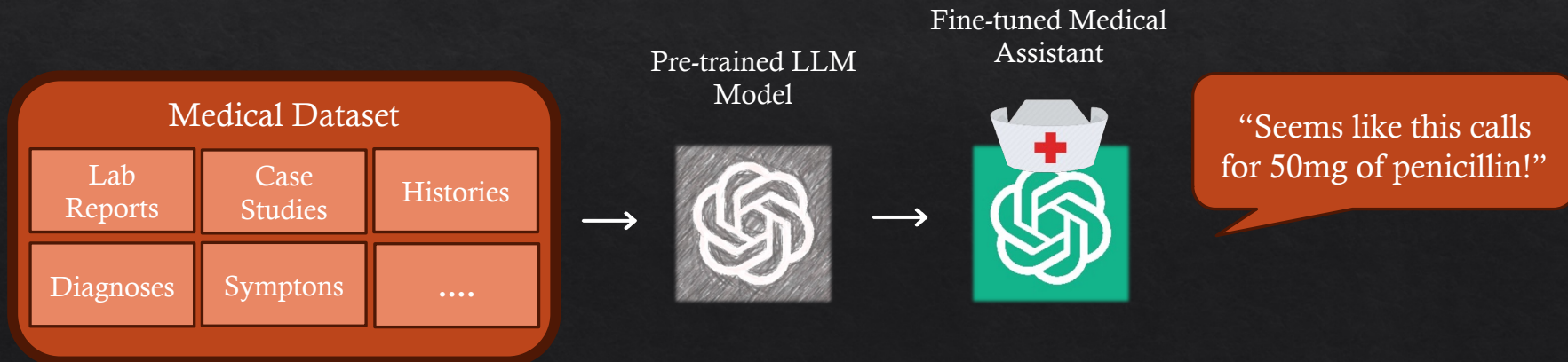


Large Language Models (LLMs)

◇ 2. Fine-tuning:

- ◇ Adapting the model to a specific domain or task.
- ◇ Using the learned parameters of the pre-trained model as a starting point, retrain the model on a (usually much smaller) domain-specific dataset. For example, LLMs for diagnosis:

Fine-Tuning



LLMs for Defensive Cybersecurity

◆ Risk Management:

- ◆ Analyze diverse data sources to identify potential risks that human analysts may overlook.
- ◆ LLMs can overcome traditional risk management challenges by analysing unstructured, siloed data, improving risk modelling, mitigation, and reporting across departments.

◆ Protection:

- ◆ Provide support and assistance during cybersecurity awareness and training.
- ◆ Automated vulnerability fixing: evaluating source code, generating corrected versions, and re-evaluating the updated code to ensure it meets security and safety requirements.
- ◆ LLMs for building attack scenarios that might be missed by internal experts.

◆ Detection:

- ◆ Perform malware detection, alert aggregation, software vulnerability detection, and phishing website detection, by fine-tuning on unstructured and semi-structured open-source cyber threat intelligence data.

References

◇ Malware:

- ◇ https://vision.ece.ucsb.edu/sites/default/files/publications/nataraj_vizsec_2011_paper.pdf
- ◇ <https://www.sciencedirect.com/science/article/pii/S1084804519303868#sec4>
- ◇ <https://www.kaggle.com/code/gvyshnya/malimg-coatnet-model>

◇ LLMs

- ◇ <https://arxiv.org/pdf/2402.00891>